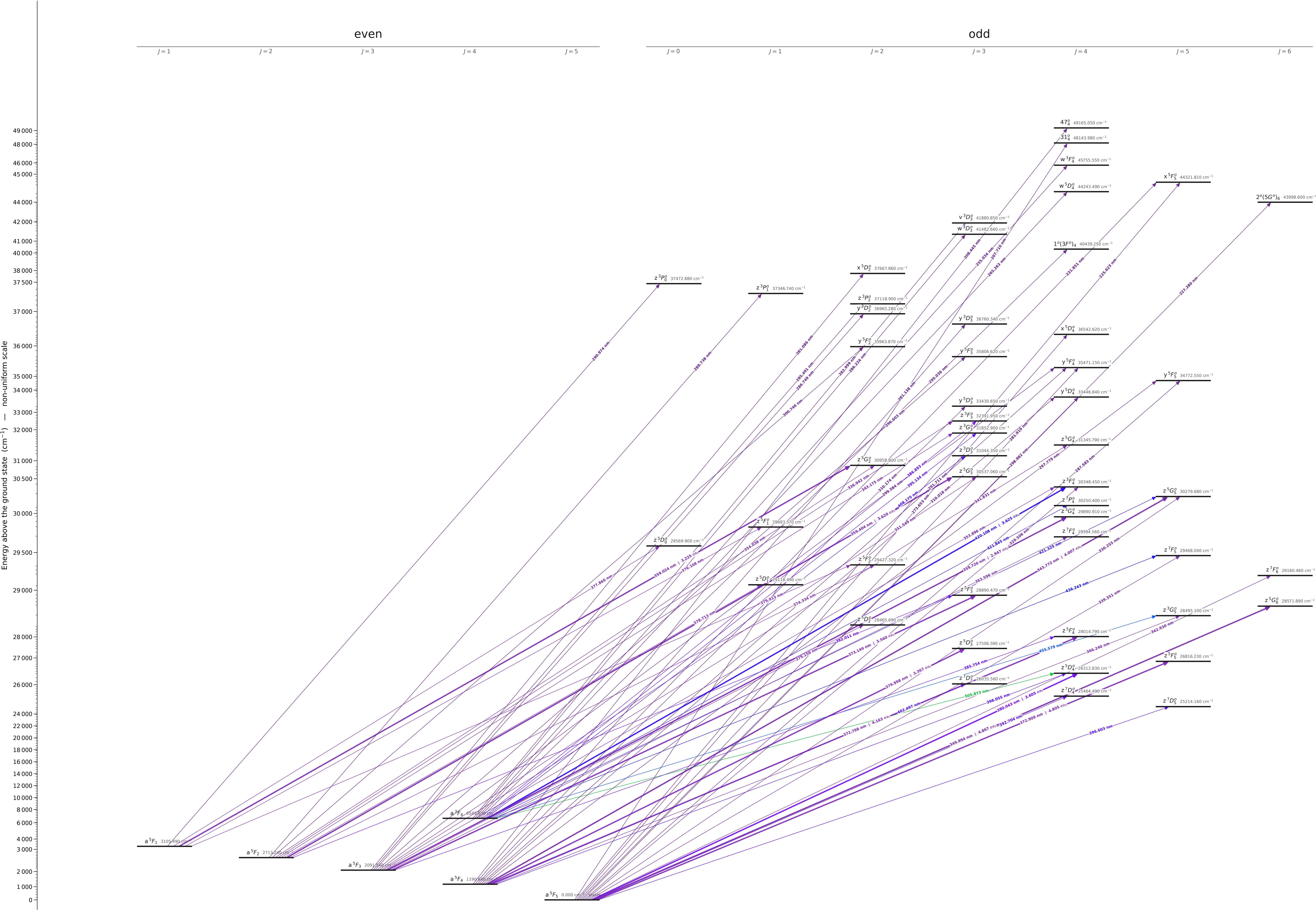
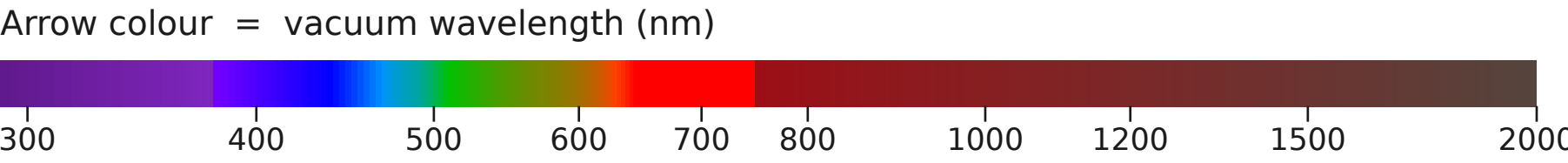


Ruthenium

Ru (Ru I). Strongest classified transitions below 2 μm (NIST): 56 levels, 70 lines.

Ru⁺ ionisation limit 59366.40 cm⁻¹ = 7.36050 eV (168.445 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(J||e r||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J e r J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁵ F ₄ - z ² G ^o ₂	343.7721	343.6736	872.0675	4.007	7.280e+07	-	-	
a ⁵ F ₅ - z ² G ^o ₂	349.9943	349.8942	856.5637	4.867	8.610e+07	-	-	
a ³ F ₁ - z ² G ^o ₂	359.0238	358.9213	835.0212	3.225	9.110e+07	-	-	
a ³ F ₂ - z ² G ^o ₂	359.4043	359.3017	834.1371	3.620	8.170e+07	-	-	
a ³ F ₃ - z ² G ^o ₂	359.7204	359.6178	833.4041	2.947	4.200e+07	-	-	
a ⁵ F ₄ - z ² F ^o ₂	372.7984	372.6924	804.1678	4.163	7.530e+07	-	-	
a ⁵ F ₅ - z ² F ^o ₂	372.9085	372.8025	803.9303	4.805	8.200e+07	-	-	
a ⁵ F ₃ - z ² F ^o ₂	373.1492	373.0431	803.4117	3.560	7.060e+07	-	-	
a ⁵ F ₄ - z ² D ^o ₂	379.9977	379.8898	788.9323	3.367	5.980e+07	-	-	
a ⁵ F ₅ - z ² D ^o ₂	380.0427	379.9349	788.8388	3.605	5.330e+07	-	-	
a ³ F ₄ - z ² F ^o ₂	420.1077	419.9894	713.6086	3.625	3.990e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).